1. HRM (Human Resource Management)

Concept: HRM means managing the people of a company — hiring them, training them, paying them, and keeping them happy so they don't leave.

Why it matters in IT: A software company's only real asset is its people. If good engineers leave, projects die. So HRM is about making sure skilled people join and stay.

Main jobs of HRM:

- Recruitment (finding right people)
- Training (upgrading skills)
- Performance reviews (tracking how they work)
- Salary and benefits (paying fairly)
- Handling conflicts
- Making sure the company follows labor laws

One-line takeaway: HRM = the function that turns a collection of individuals into a working, loyal team.

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2. Software Ethics Under Deadline Pressure

Concept: Sometimes managers push engineers to ship buggy software just to meet a deadline. The question is: should you agree or refuse?

The rule: Safety always beats deadlines. Especially in healthcare, aviation, banking, or anything where bugs can kill or harm people.

What an ethical engineer does:

1. Document the bug in writing
2. Tell senior management clearly about the risk
3. Refuse to release unsafe software

4. Suggest a backup plan (delay, partial release, etc.)

Famous real case: Therac-25 machine (1985) gave patients massive radiation overdoses because engineers ignored a known bug under deadline pressure. 3 people died.

One-line takeaway: Never ship software you know is unsafe, no matter who pressures you.

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3. Client Contract (for Software)

Concept: A contract is a written agreement between the software company and the client that says exactly what will be built, when, for how much, and who owns what.

Why it's important: Software is invisible. Without a contract, clients and vendors fight over "you didn't give me what I asked for."

Main things every software contract must cover:

- What the software will do (modules, features)
- Documentation (user manual, technical docs)
- Testing (how we prove it works)
- Deployment and training
- Who owns the source code
- Payment schedule
- Deadlines
- Confidentiality (NDA)
- What happens if things go wrong (warranty, termination)

One-line takeaway: A good contract prevents fights. A vague contract causes disasters (like the Healthcare.gov site that crashed on launch).

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4. IT Company Structures

Concept: "Structure" = how a company is organized. Who reports to whom, who makes decisions.

Legal forms (how the business exists in law):

- **Sole trader:** one person, no paperwork, but personally liable for everything
- **Partnership:** two or more people, still personally liable
- **Limited company:** separate legal entity, owners' liability limited to what they invested (this is what most software houses are)

Internal structures (how teams are arranged):

- **Functional:** grouped by skill (all devs in one team, all QA in another)
- **Divisional:** grouped by product (Azure team, Office team at Microsoft)
- **Matrix:** you have two bosses, a skills manager and a project manager
- **Flat:** very few managers, fast decisions (startups)

Software house hierarchy: Managing Director on top, with Technical, Sales, Operations, Financial, and Overseas directors below.

One-line takeaway: Structure decides speed and clarity. Wrong structure = bottlenecks and confusion.

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5. Ethical Culture in Software Companies

Concept: Ethical culture = the unwritten rules of how people actually behave at work. Do they hide bugs? Bully juniors? Or speak up and support each other?

Why it matters: Engineers work alone on code. No one watches every line. So ethics has to be built into the culture, not forced by managers.

6 things a company does to build ethical culture:

1. Write a clear Code of Conduct

2. Protect work-life balance (no forced overtime)
3. Allow safe feedback (anonymous hotlines)
4. Reward ethical behavior, not just output
5. Managers must lead by example
6. Regular ethics training

Bad example: Uber under Travis Kalanick — harassment, ignoring rules, "hustle at any cost." Ended in scandal.

Good example: Microsoft under Satya Nadella changed from "know-it-all" to "learn-it-all" culture. Stock went from 400.

One-line takeaway: Culture is what people do when the boss is not looking.

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6. Career Growth and Retention

Concept: Retention = keeping employees from leaving. Career growth = giving them a path to move up. The two are linked.

Why it matters: Replacing an engineer costs 1.5 to 2 times their salary. So companies fight to keep good people.

7 things companies do for long-term careers:

1. Understand what each employee wants (manager track or technical track?)
2. Offer clear promotion paths
3. Provide training, courses, certifications
4. Succession planning (train juniors for senior roles)
5. Prevent burnout (work-life balance)
6. Give honest, regular feedback
7. Align personal growth with company goals

Example: Google has clear level-based ladders (L3 to L10). Netflix pays top-of-market but has a "Keeper Test" — managers ask "would I fight to keep this person?"

One-line takeaway: People stay where they grow. No growth = they leave.

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7. Intellectual Property (IP) in Software

Concept: IP = legal ownership of ideas and creations. In software, it protects your code, algorithms, designs, and brand from being copied.

4 types of IP (memorize these):

Type	Protects	Example
Copyright	Original expression (source code, UI, docs)	Windows source code
Patent	New inventions/algorithms (20 years)	Amazon's 1-Click ordering
Trademark	Brand identity (names, logos)	Apple logo, "iPhone" name
Trade Secret	Confidential info kept secret	Coca-Cola formula, Google search algorithm

For a patent, invention must be: (1) new, (2) non-obvious, (3) industrially useful.

Important: Patents are country-specific. No worldwide patent exists.

In Pakistan: Copyright Ordinance 1962, Patents Ordinance 2000, Trade Marks Ordinance 2001. WIPO (UN agency in Geneva) handles it globally.

Why IP protection matters:

1. Rewards creators
2. Encourages R&D investment
3. Protects consumers from fakes
4. Helps tech transfer to developing countries

One-line takeaway: IP turns intangible ideas into legal property you can sell, license, or defend.

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8. Trade Secret Protection

Concept: A trade secret is confidential business info that gives you an advantage — like a secret algorithm or client list. You keep it secret instead of patenting it.

Why use a trade secret instead of a patent?

- Lasts forever if kept secret (Coca-Cola formula is 135+ years old)
- No registration needed
- Patents require you to publish the secret — trade secrets don't

Legal test: For the law to protect you, you must take "reasonable steps" to keep it secret.

4 main ways to protect trade secrets:

1. **NDAs** — everyone who sees it signs a non-disclosure agreement
2. **Access control** — only a few people can see the full secret
3. **Employment contracts** — ex-employees cannot share it after leaving
4. **Documentation** — keep records proving you took protection seriously

Famous case: Anthony Levandowski stole 14,000 files from Google's Waymo, took them to Uber. Uber paid $245 million settlement. He went to prison.

One-line takeaway: Trade secrets last forever, but only if you actually keep them secret.

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9. IEEE and ACM Code of Ethics

Concept: These are two international rulebooks for how computer professionals should behave. Like doctors have the Hippocratic Oath.

ACM: Association for Computing Machinery. Main computer society.**IEEE:** Institute of Electrical and Electronics Engineers. Main engineering society.

Together they made the Software Engineering Code with 8 principles. Each principle covers one stakeholder:

1. **Public** — act in public interest
2. **Client and Employer** — serve them, but not against public interest
3. **Product** — deliver high quality
4. **Judgment** — stay independent, no bribes
5. **Management** — managers must promote ethics
6. **Profession** — protect the reputation of software engineering
7. **Colleagues** — be fair to teammates
8. **Self** — keep learning and follow the code

Easy trick to remember: P-C-P-J-M-P-C-S

Key rule across both codes: Public safety comes FIRST. Always.

Violation example: Volkswagen engineers wrote code to cheat emissions tests. Broke "Public," "Product," "Judgment," and "Profession" principles. VW paid $30 billion in fines. Some engineers went to jail.

One-line takeaway: The code is your backup when the boss asks you to do something wrong. "I can't do this, it violates my professional code."

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10. Social Computing

Concept: Social computing = using computers for social stuff. Social networks, blogs, video sharing, wikis, crowdfunding. Basically, any online thing where people interact with other people.

Main examples:

- Social networks: Facebook, Instagram, LinkedIn, X

- Blogs: Medium, Substack
- Video: YouTube, TikTok
- Wikis: Wikipedia
- Crowdfunding: Kickstarter, GoFundMe
- Reviews: Amazon, eBay

Ethical problems it creates:

1. **Privacy loss** — your data is collected and sold
2. **Misinformation** — fake news spreads fast
3. **Cyberbullying** — anonymity encourages cruelty
4. **Addiction** — especially affects teen mental health
5. **Copyright violations** — people share stuff they don't own
6. **Echo chambers** — you only see views you already agree with

James Moor's Invisibility Factor (important concept): Computers do 3 hidden things:

1. Invisible abuse (hidden data collection)
2. Invisible programmer bias (biased algorithms)
3. Invisible calculations (you can't verify what AI decides about you)

Famous bad case: Cambridge Analytica (2018) harvested 87 million Facebook users' data to target political ads. Facebook fined $5 billion.

Famous good case: Arab Spring (2011) — Facebook and Twitter helped people organize revolutions against dictators.

One-line takeaway: Social computing connects humanity but also creates new ethical problems law hasn't caught up with yet.

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11. Professional Responsibility of a Software Engineer

Concept: A software engineer is not just a coder. Like a doctor or lawyer, they have duties to the public, client, profession, and themselves. This is called "professional responsibility."

Why software engineers have extra responsibility:

- Software runs hospitals, planes, banks, power grids
- Bugs can kill people or destroy money
- Users can't tell if code is safe — they trust engineers
- Engineers know risks that managers often don't

The 8 duties (same as ACM/IEEE code above):

1. Duty to public (safety first)
2. Duty to client and employer
3. Duty to deliver a good product
4. Duty of honest judgment
5. Duty in management (if you're a lead)
6. Duty to the profession
7. Duty to colleagues
8. Duty to self (keep learning)

Responsibility vs. Liability:

- **Liability** = what the law forces on you
- **Responsibility** = what ethics asks from you (broader than law)

What a responsible engineer looks like:

- Refuses to ship unsafe code
- Only takes on work they're qualified for
- Writes honest documentation
- Credits others' work
- Keeps learning throughout career
- Speaks up when something is wrong

Famous failure: Therac-25 engineers knew about the bug but stayed quiet. 3 patients died. Every duty was violated.

One-line takeaway: Being a professional means doing the right thing, even when nobody is forcing you to.

Quick Exam Writing Formula

For ANY question, write:

1. **Definition** — one clear sentence
2. **Why it matters** — one line
3. **Key points** — 4 to 6 bullets
4. **One real example** — shows you understand

That gets you full marks. Good luck, you've got this.